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Professor Alán Aspuru-Guzik
Editor-in-Chief, *Digital Discovery*
Royal Society of Chemistry
Thomas Graham House, Science Park
Milton Road, Cambridge CB4 0WF, UK

Dear Professor Aspuru-Guzik,

We are pleased to submit our manuscript entitled **“How Far Can Structure-Based Machine Learning Predict Experimental Singlet–Triplet Gaps? An Honest Benchmark for TADF Emitter Triage”** for consideration as a Research Article in *Digital Discovery*.

Why *Digital Discovery*? The journal has been a leading venue for rigorous, reproducible, and self-critical machine learning in chemistry—including negative and cautionary results. Our manuscript is in that spirit: a carefully validated assessment of what cheap, structure-based ML can and cannot do for a property of central interest, with every number reproducible from deposited code.

Scientific contributions. The singlet–triplet gap (ΔE_{ST}) governs thermally activated delayed fluorescence (TADF) emitters. We make four contributions:

1. **A diagnosis of feature–target leakage.** We show that regressing ΔE_{ST} onto feature sets containing the S_1 and T_1 energies is circular ($\Delta E_{ST} \equiv E_{S_1} - E_{T_1}$); reported sub-0.05 eV accuracies for this set-up reflect arithmetic, not learning—a general leakage failure mode for any property defined as a difference of computed quantities.
2. **A scaffold-validated benchmark.** On 231 molecules across 212 Bemis–Murcko scaffolds, a random forest on energy-free descriptors predicts *experimental* ΔE_{ST} with MAE = 0.096 eV ($R^2 = 0.25$, $\rho = 0.36$); a label-permutation test ($p = 0.001$) confirms the signal is real. Structural descriptors and fingerprints perform equivalently, and the semi-empirical excited-state step adds nothing.
3. **An explicit accuracy ceiling.** Inter-laboratory scatter (0.1–0.15 eV) and functional dependence of the reference (up to 0.58 eV between B3LYP and CAM-B3LYP) bound achievable accuracy near 0.1 eV; a vertical-versus-adiabatic check (CAM-B3LYP/def2-TZVP) shows the cheap descriptors carry only a systematic, rank-preserving offset. The result has direct implications for how such models should be reported.
4. **A practical triage filter** that enriches low-gap candidates by $1.2\text{--}1.4\times$ over random—useful for prioritisation, not a device-level or emission-spectrum predictor.

Novelty and differentiation. This work is distinct from our two prior *JCIM* publications (xTB/sTDA benchmarking and design-rule extraction): here we ask, rigorously, what cheap

structural data can and cannot support for ΔE_{ST} prediction, and we report the limitations openly.

Open science. All computational data, trained models, SHAP analysis, and Python code are openly available on Zenodo (DOI: 10.5281/zenodo.17436069) under CC-BY-4.0 license, ensuring full reproducibility and enabling extension to new chemical spaces.

APC waiver. As corresponding author affiliated with the University of Yaoundé I, Cameroon, we are eligible for a 100% APC waiver under the RSC's Research4Life programme for authors from low- and middle-income countries. We will provide the required documentation upon request.

Suggested reviewers. We suggest the following experts who have the relevant background in ML for materials discovery and TADF photophysics:

- Prof. Heather Kulik (MIT) — ML for molecular design, active learning
- Prof. Andrew Monkman (Durham) — TADF photophysics, device performance
- Prof. Olexandr Isayev (Carnegie Mellon) — ML for chemistry
- Prof. Denis Jacquemin (Nantes) — TD-DFT for excited states

We confirm that this manuscript has not been published elsewhere and is not under consideration by any other journal. All authors have approved the manuscript and agree to its submission to *Digital Discovery*.

We hope you will find this work suitable for publication and look forward to your response.

Yours sincerely,

Jean-Pierre Tchapet Njafa
(on behalf of all authors)